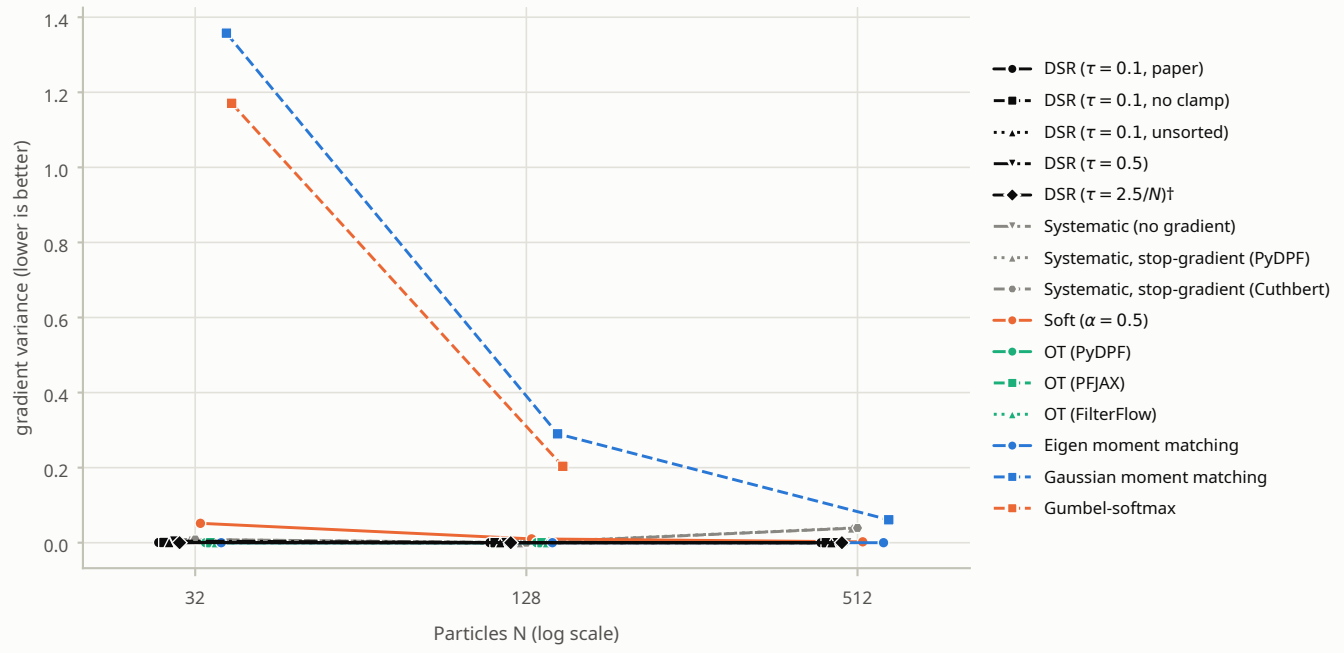
gradient variance (lower is better)
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
32
128
512
Particles N (log scale)
DSR (τ = 0.1, paper)
DSR (τ = 0.1, no clamp)
DSR (τ = 0.1, unsorted)
DSR (τ = 0.5)
DSR (τ = 2.5/N)†
Systematic (no gradient)
Systematic, stop-gradient (PyDPF)
Systematic, stop-gradient (Cuthbert)
Soft (α = 0.5)
OT (PyDPF)
OT (PFJAX)
OT (FilterFlow)
Eigen moment matching
Gaussian moment matching
Gumbel-softmax